Cover Note – HybridDesk Employee Rostering Application

Project Overview

As part of the **ICAI AICA Level 2 Capstone Project, Batch 80**, I have developed **HybridDesk – an Employee Rostering Application** designed to address the practical challenges of employee rostering in a hybrid working environment.

The COVID-19 pandemic significantly changed workplace practices and accelerated the adoption of hybrid working models. In a Global Investment Bank environment, where hot-desking is widely used and available seating is often lower than total employee headcount, effective employee rostering becomes a complex operational challenge. The difficulty increases further when employees are required to meet role-based Work From Office (WFO) compliance requirements while accounting for limited seating capacity, planned absences, unexpected absences, public holidays, business travel and exceptional leave scenarios.

HybridDesk was developed to address this challenge by creating **optimized employee roster plans** while simultaneously providing **management-level compliance and capacity metrics** for decision-making and reporting.

Problem Statement

The core problem addressed by the application is the **efficient allocation of employees to office seats while maintaining WFO policy compliance**.

The application considers different minimum WFO requirements based on employee designation, such as:

- MDs and EDs – minimum 80% WFO compliance
- VPs – minimum 70% WFO compliance
- Professionals – minimum 60% WFO compliance

These requirements must be managed within constrained seating capacity and while considering employee availability, planned or unplanned absences, public holidays, work travel and exceptional situations such as maternity/paternity leave and compensatory offs.

Managing these variables manually through spreadsheets or conventional rostering methods can be time-consuming and may make it difficult to identify the optimal allocation. HybridDesk addresses this by providing a structured, automated approach to employee rostering.

Application Objectives

The key objectives of HybridDesk are to:

1. Generate optimized employee roster plans for a two-week period.
2. Ensure employee allocations consider WFO policy requirements and available seating capacity.
3. Incorporate employee availability, absences, holidays, travel and exceptional scenarios.
4. Provide executive-level metrics for monitoring compliance and capacity utilization.
5. Enable users to assess the impact of changing business assumptions through a real-time **What-if Simulator**.

Key Features

HybridDesk generates optimized rosters for the upcoming fortnight and provides management-oriented metrics to support decision-making.

A key feature is the **What-if Simulator**, which allows users to dynamically change assumptions such as available seat capacity and required compliance percentages. The application immediately reflects the resulting impact on compliance levels and seat deficits, helping management understand capacity and policy trade-offs.

The application is organized into multiple modules covering employee and rostering-related information, with the demonstration using an employee master containing 25 employees.

Technology and Development Stack

The project was developed using an **AI-assisted application development approach**.

The documented development stack and methodology were:

- **Vibe Coding** – used to define and structure the business and functional requirements for optimal seat allocation.
- **Claude Opus 5 through the ICAI AI Agent** – used to convert the business requirements into a structured application-building prompt.
- **Google AI Studio** – used as the application development environment, where the structured prompt was used to build the HybridDesk application.

The project therefore demonstrates how **generative AI can be combined with domain knowledge and business-process understanding to rapidly develop a practical business application**.

The uploaded project materials do not explicitly identify a specific underlying programming language or software framework; therefore, the technology stack above reflects the development tools and methodology documented for the project rather than assumptions about the application's underlying codebase.

Business Value

HybridDesk demonstrates how AI-enabled application development can be applied to a real-world workforce planning problem. By combining business rules, employee availability, seating constraints and compliance requirements into a single solution, the application can reduce manual effort, improve roster planning and provide management with greater visibility into compliance and capacity constraints.

The project also demonstrates the practical application of **Artificial Intelligence, prompt engineering, automation and business-domain expertise** to solve an operational problem in a structured and scalable manner.

Conclusion

HybridDesk is a practical demonstration of how AI can be leveraged to transform a traditionally manual employee rostering process into a more intelligent, data-driven and interactive solution. The project combines domain knowledge of hybrid working and compliance requirements with AI-assisted application development to create a solution focused on **optimal seat allocation, policy compliance and management decision support**.